

Spectral diagnostics and local fixed-budget sensitivity of critical points in finite encounter-reaction models

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Reaction-time densities of confined encounter reactions can contain several transport-generated time windows, yet whether those windows appear as critical points of the measured flux depends jointly on transport, the initial law, and the spatial reaction field. We develop an exact finite-model calculus for diagnosing and steering those critical points. First, a reaction-support Green reduction separates free transport from volume killing. Second, the generalized Descartes rule gives a necessary spectral gate: a reversible finite-model density with m interior modes requires at least $2m - 1$ sign changes in its ordered nonzero residues. Third, an exact Fréchet–Duhamel derivative gives the response of the critical-point equation to every killing rate, and projection onto the tangent space of a declared reactivity budget identifies the locally steepest feasible redistribution in a chosen metric.

Two families of finite generators exercise the calculus. Direct matrix-exponential derivatives locate a nondegenerate reaction-density fold in a finite two-reactant continuous-time Markov chain under a rate-ratio control, and independent continuation recovers the generic $1/2$ critical-point-separation and $3/2$ prominence exponents of the fold normal form. A separate finite-radius two-dimensional family redistributes reactivity at an exactly fixed per-grid killing sum and exhibits the same nondegenerate fold structure and exponents on three grids. The critical controls of the two-dimensional family are grid- and budget-measure sensitive and are reported as finite-generator mechanism certificates. An exact fold-transfer theory, a cell-averaged refinement ladder run under two independent transport discretizations, and a 2×10^7 -walker lattice-free Brownian realization then settle the continuum question: the physical matched-budget path carries no interior continuum fold; the finite-grid critical controls are coarse-discretization artifacts, and the organizing fold is extrapolated to the admissibility edge of the family, where the far-channel rate vanishes. The fold-transfer proof, the refinement-ladder roots, and detailed endpoint, coordinate, clock, capacity, and multichannel audits are reported in the Supplemental Material.

I. INTRODUCTION

The reaction time of two mobile particles can retain information that a mean reaction time or a single long-time decay rate suppresses. In confinement, first contact need not be reaction: encounter can be imperfect and reaction can be restricted to selected spatial regions. The measured density then depends jointly on transport, encounter geometry, and the reaction observable. Multiple critical points of the density—time-stationary points in the sense of singularity theory, not critical phenomena—can reflect dynamically distinct reaction streams, but their existence does not by itself identify which mechanism produced them.

Multimodal passage and reaction-time laws have established transport, geometric, internal-state, and encounter-history mechanisms. Confined encounter laws, exact propagators, heterogeneous transmission sites, multi-target reductions, and transport-generated multiple peaks are developed in Refs. [1–9]. Imperfect reaction has likewise been treated through Doi volume killing, radiation boundaries, and heterogeneous reactivity [10–15]. Static killing fields and partially reactive network elements have already been rearranged at

fixed total strength, and position-dependent Robin reactivity has been optimized under an integral constraint [16–18]. Besga *et al.* previously established a Brownian first-passage shape transition in which a finite-time maximum–minimum pair merges under a target-distance/initial-spread control [19]. We therefore do not claim either the fold criterion $f_t = f_{tt} = 0$, a two-clock shape transition, or any of the other ingredients above as new.

The narrower problem here is differential. Suppose a finite killed generator has a reaction-density fold, so that $f_t = f_{tt} = 0$ at positive time. Can one (i) determine whether its observable spectrum has enough sign variation to permit the requested number of modes, (ii) compute the response of the fold equation to a spatial killing redistribution, and (iii) identify the most effective infinitesimal redistribution compatible with a fixed budget? We answer those questions exactly for declared finite generators, then test the resulting calculus in a finite encounter chain and a finite-radius two-dimensional family.

The distinction between a *strict extremum* and a *classifier-resolved mode* is important. The main article uses derivative-certified folds only. Endpoint morphology, synchronous-clock controls, catalytic-coordinate sensitivity, finite-grid trimodality, capacity calibration, and generalized-inverse-Gaussian (GIG) screening are

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preserved as secondary audits in the standalone Supplemental Material [20]. Those calculations delimit the causal language but are not needed to establish the two fold certificates presented here.

The seven main claims and their boundaries are summarized in Table I; a transfer theory, a cell-averaged refinement ladder, and an off-lattice realization (Sec. V) settle what the finite certificates do and do not imply in the continuum. Figure 1 shows the spatial channel picture and the generic fold geometry.

II. FINITE ENCOUNTER MODEL AND GREEN REDUCTION

Let $X_i(t) \in \Omega \subset \mathbb{R}^d$ be two independent particles before reaction, with generator $\mathcal{L}_0$ and reflecting outer boundaries. A finite contact radius $a > 0$ and an affine catalytic coordinate

$$C_\eta(x_1, x_2) = \eta x_1 + (1 - \eta)x_2, \quad 0 \leq \eta \leq 1,$$

define channel killing fields

$$\begin{aligned} K_{a,j}^{(\eta)} &= \kappa_j(C_\eta) \mathbf{1}_{C_j}(C_\eta) \mathbf{1}_{\{|x_1 - x_2| < a\}}, \\ K_a^{(\eta)} &= \sum_j K_{a,j}^{(\eta)}. \end{aligned} \quad (1)$$

The unreacted joint density obeys

$$\partial_t p = \mathcal{L}_0 p - K_a^{(\eta)} p. \quad (2)$$

With $S(t) = \int p$ and $f_j(t) = \int K_{a,j}^{(\eta)} p$, reflecting boundaries give the exact mass balance

$$-S'(t) = f(t) = \sum_j f_j(t). \quad (3)$$

The catalyst coordinate is part of the physical model: the midpoint and the diffusion-decoupling weighted centre are unequal when the particle diffusivities differ. Coordinate algebra, the transformed domain, Doi–Robin qualifications, and the finite-grid midpoint audit are given in the Supplemental Material [20].

Write the volume killing operator on $\mathbf{H} = L^2(\mathcal{D})$ as

$$V = \Gamma^* K \Gamma, \quad \mathcal{T} = \mathcal{L}_0 - V, \quad (4)$$

where Γ restricts a function to the reaction support, Γ^* extends it by zero, and K is a bounded nonnegative multiplication operator on that support. It may have a kernel. For $\text{Re } s$ above the spectral bound, define

$$\mathcal{R}_0(s) = (s - \mathcal{L}_0)^{-1}, \quad \mathcal{G}(s) = \Gamma \mathcal{R}_0(s) \Gamma^*. \quad (5)$$

Whenever the displayed inverse exists, the inverse-free resolvent identity is

$$(s - \mathcal{T})^{-1} = \mathcal{R}_0 - \mathcal{R}_0 \Gamma^* K (I + \mathcal{G}K)^{-1} \Gamma \mathcal{R}_0. \quad (6)$$

Thus, for $u = \Gamma \mathcal{R}_0 q_0$, the restricted state and transformed reaction density are

$$x = (I + \mathcal{G}K)^{-1}u, \quad y = K(I + \mathcal{G}K)^{-1}u. \quad (7)$$

No inverse of K is required. This separation exposes free transport through $\mathcal{G}$, geometry through Γ , and intrinsic chemistry through K .

For a finite continuous-time Markov chain (CTMC), adopt row vectors. Let $L_0 = L_1 \oplus L_2$, let columns of U select reactive states, and let K be diagonal in channel rates. Then

$$T = L_0 - UKU^\top, \quad f_j(t) = \alpha e^{Tt} UK e_j. \quad (8)$$

The finite support matrix

$$G(s) = U^\top (sI - L_0)^{-1} U$$

gives

$$\det(sI - T) = \det(sI - L_0) \det[I + G(s)K].$$

The second determinant locates only candidate killed poles coupled to the reaction support. Dark transport modes, numerator cancellation, and observable residues must still be checked. Splitting probabilities are computed from the stable killed solve

$$\pi = \alpha(-T)^{-1}UK, \quad (9)$$

when the reachable live chain is transient. Exact time derivatives are

$$\partial_t^n f(t) = \alpha e^{Tt} T^n UK \mathbf{1}. \quad (10)$$

The detailed two-site exclusion test model and zero-frequency diagnostics are reported in the Supplemental Material [20].

III. SPECTRAL GATE AND FIXED-BUDGET SUSCEPTIBILITY

A. A necessary sign-variation condition

Consider a finite killed model whose observable density has a real, diagonalizable spectral expansion. After equal decay rates are grouped and zero residues removed, write

$$f(t) = \sum_{j=1}^n a_j e^{-\lambda_j t}, \quad 0 < \lambda_1 < \dots < \lambda_n. \quad (11)$$

Reversibility is a sufficient hypothesis because detailed balance makes the killed generator similar to a real symmetric matrix. Applying the classical generalized Descartes rule [21] to

$$f_t(t) = - \sum_{j=1}^n \lambda_j a_j e^{-\lambda_j t} \quad (12)$$

TABLE I. Evidence hierarchy used in the main article.

Statement	Evidence	Boundary
Reaction-support Green reduction	Exact conditional operator identity and finite Woodbury specialization	Volume killing on a declared support; no unqualified Robin trace identity or continuum meromorphic continuation
Finite reversible spectral gate	Generalized-Descartes corollary and primary-generator reconstruction	Necessary, not sufficient; not a channel-rank bound and not applied to complex-spectrum, Jordan, or continuum laws
Fixed-budget modality susceptibility	Exact finite Fréchet–Duhamel identity, projected optimum, and independent directional checks	Infinitesimal interior-rate design in a declared budget and metric; no finite-amplitude, binary-mask, moving-support, or continuum optimum
Finite encounter-chain fold	Direct finite-matrix derivatives, root solve, residuals, transversality, and independent normal-form scaling	One finite model; minority-channel splitting probability 8.95×10^{-6}
Finite-radius 2D matched-budget folds	Three finite-generator folds, independent normal-form scaling, and a second budget-measure check	Critical controls are nonmonotone and budget-sensitive; no converged continuum critical control
Fold-transfer theory	Exact transfer statements with Supplemental Material proof	Sectorial and jet hypotheses instantiated numerically, not proved for these schemes
Continuum bridge (negative result)	Cell-averaged two-discretization ladder to 33^2 nodes; 2×10^7 -walker off-lattice realization	Negative result for this geometry only; sectorial and jet hypotheses instantiated numerically; five sampled off-lattice controls; extrapolants are estimates with the cross-scheme gap reported alongside

after $x = e^{-t}$ gives the following corollary:

$$\#\{\text{positive-time zeros of } f_t\} \leq V(a_1, \dots, a_n),$$

where zeros are counted with multiplicity and V is the number of ordered residue sign changes. Hence m non-degenerate interior maxima separated by minima require

$$V(a_1, \dots, a_n) \geq 2m - 1. \quad (13)$$

This is a necessary falsification gate, not a mode-count predictor. Three sign changes can coexist with only one positive critical point, and a rank-one killing support can already produce a bimodal phase-type density. Reaction-support rank therefore cannot replace the observable residues.

Full symmetric eigendecompositions rebuild the two primary generators used in the numerical audits. Reversibility, the gate’s hypothesis, is verified directly for both free generators: every live transition is two-way, a spanning-tree reconstruction of the detailed-balance log-weights closes on all off-tree edges with maximum cycle inconsistency 2.4×10^{-14} (961 states) and 9.1×10^{-14} (2025 states), and the recovered measure symmetrizes the killed generator. Equal decay rates are grouped at a relative tolerance 2×10^{-10} before residues are summed; at that tolerance no two eigenvalues of either generator merge. The 961-state finite encounter generator is evaluated at its three reported critical points; the separate 2025-state three-channel generator is evaluated at five detected critical points. Across the four relative residue

cutoffs 10^{-12} , 10^{-10} , 10^{-8} , and 10^{-6} (relative to the summed absolute grouped residues), the minimum retained sign counts are 444 and 191, respectively, above the necessary bounds three and five. Because these reconstructions reuse the same generator constructors and critical-time artifacts, they test decomposition and reconstruction rather than discover roots independently. Full cutoff, counterexample, and reconstruction details are in the Supplemental Material [20].

B. Budget-projected modality response

For a finite live-state transport generator L , let

$$T(k) = L - \text{diag } k, \quad f(t; k) = \alpha e^{T(k)t} k, \quad (14)$$

where $k > 0$ on the allowed design support. For $k_\epsilon = k + \epsilon h$, put $H = -\text{diag } h$. The exact Fréchet–Duhamel identity is

$$\text{De}^{Tt}[H] = \int_0^t e^{T(t-s)} H e^{Ts} ds \quad (15)$$

[22]. Since $f_t = \alpha e^{Tt} T k$, its complete directional derivative is

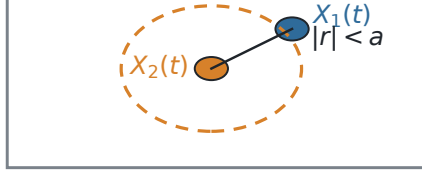
$$D_k f_t(t; k)[h] = \alpha \text{De}^{Tt}[H] T k + \alpha e^{Tt} (H k + T h). \quad (16)$$

The second term is essential because the killing field is also the measured reaction observable. Writing this lin-

Spatial encounter channels and their modality boundary

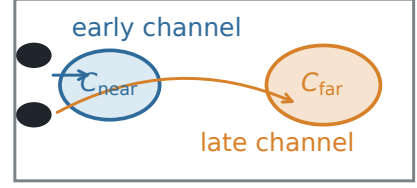
(a) Finite-radius encounter

$$\begin{aligned} r &= X_1 - X_2 \\ R &= (D_2 X_1 + D_1 X_2) / (D_1 + D_2) \\ C(\eta) &= \eta X_1 + (1 - \eta) X_2; \text{ bounded 2D: } \eta = 1/2 \end{aligned}$$



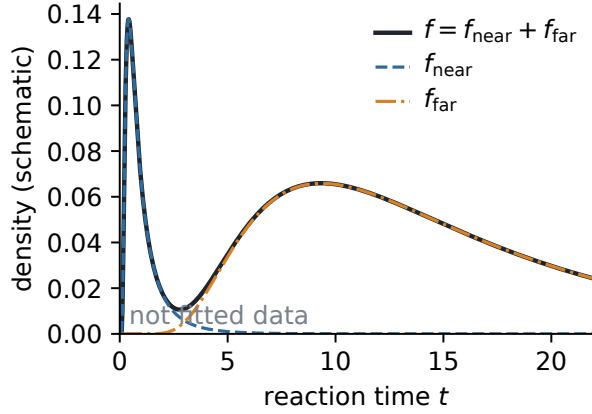
Reflecting domain; contact + catalyst required

(b) Pattern selects streams



$$K_a(\eta) = \sum_j \kappa_j \mathbb{1}[C_j](C(\eta)) \mathbb{1}[|r| < a]$$

(c) Channel mixture



(d) Generic modality fold

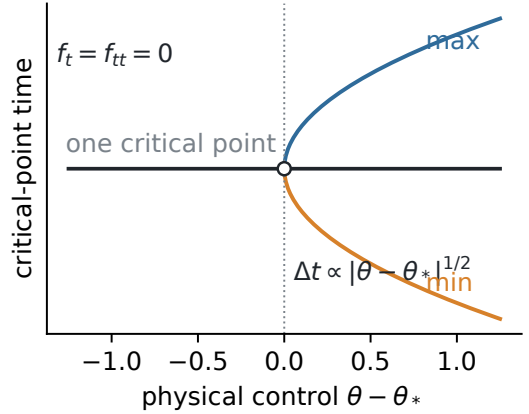


FIG. 1. Spatial encounter channels and their modality boundary. (a) Reaction requires finite-radius contact and catalytic activation at a declared affine encounter location. (b) Separated patches select early and late streams. (c) Their sum can be bimodal; the curves are schematic and are not fitted data. (d) At a generic fold, a maximum–minimum pair is created with square-root separation under a physical control.

ear functional as $g^T h$,

$$g_i = (\alpha e^{Tt} T)_i - k_i (\alpha e^{Tt})_i - \int_0^t (\alpha e^{T(t-s)})_i (e^{Ts} T k)_i ds. \quad (17)$$

Let $c > 0$ define the declared budget $c^T h = 0$, and let $M \succ 0$ define the local norm $h^T M h = 1$. Put

$$\lambda_B = \frac{c^T M^{-1} g}{c^T M^{-1} c}, \quad \tilde{g} = g - \lambda_B c. \quad (18)$$

Cauchy–Schwarz on the budget tangent space gives

$$h_* = \frac{M^{-1} \tilde{g}}{\sqrt{\tilde{g}^T M^{-1} \tilde{g}}}, \quad \max D_k f_t[h] = \sqrt{\tilde{g}^T M^{-1} \tilde{g}}. \quad (19)$$

At a double stationary point, $\tilde{g} \neq 0$ means that some infinitesimal feasible redistribution is transverse to the fold. The direction depends on the selected budget c and metric M ; it is not a coordinate-free global optimum.

An eleven-state test model, with seeded random positive diagonal metric M and budget vector c (uniform draws on $[0.65, 1.85]$ and $[0.55, 1.65]$), compares Eq. (17) with a basis Fréchet calculation, 256 independently generated budget-zero finite differences, a state permutation, and 20,000 random feasible directions. The relative Duhamel error is 3.3×10^{-15} , the maximum independent finite-difference error is 2.4×10^{-8} , and the best random response is 0.933 of Eq. (19). At the finite encounter fold below, the chain-rule reconstruction of the computed log-rate transversality agrees to 6.8×10^{-15} relative error. The orthogonal near-to-far fixed-state-sum redistribution has nonzero transversality -1.1621×10^{-5} .

IV. PHYSICAL FOLDS IN FINITE ENCOUNTER GENERATORS

At a fold of density critical points under a physical parameter θ ,

$$f_t(t_*, \theta_*) = f_{tt}(t_*, \theta_*) = 0. \quad (20)$$

The generic nondegeneracy conditions are

$$a_* = f_{t\theta}(t_*, \theta_*) \neq 0, \quad b_* = f_{ttt}(t_*, \theta_*) \neq 0. \quad (21)$$

Taylor expansion gives

$$f_t = a_* \Delta\theta + \frac{b_*}{2} \Delta t^2 + o(|\Delta\theta| + |\Delta t|^2).$$

On the side where $-2a_*\Delta\theta/b_* > 0$, the new maximum and minimum separate as

$$\Delta t_{\pm} = \pm \sqrt{-\frac{2a_*}{b_*} \Delta\theta} + o(|\Delta\theta|^{1/2}), \quad (22)$$

and integrating the quadratic derivative gives prominence proportional to $|\Delta\theta|^{3/2}$. These equations concern a physical control that can change both channel weights and shapes; a freely adjusted mixture weight is not substituted for θ .

A. Finite two-reactant CTMC certificate

The finite encounter chain has sites $0, \dots, 30$. Its two independently clocked walkers have interior total (attempt) jump rates 0.7 and 0.2, respectively, and a common right bias 0.3. Their interior left/right rates are therefore (0.245, 0.455) and (0.07, 0.13). At a boundary an out-of-range jump is omitted, giving the declared reflecting (holding) convention. If L_1 and L_2 are the corresponding row generators, the free pair generator is the Kronecker sum $L_1 \oplus L_2 = L_1 \otimes I + I \otimes L_2$, and the initial law is concentrated at (3, 15). Killing acts only at the reactive co-locations (13, 13) and (30, 30), with rates κ_{near} and κ_{far} , respectively. Transport and the initial law remain fixed. We set $\kappa_{\text{far}} = -\log 0.4 = 0.9162907319$ and vary

$$\theta = \log(\kappa_{\text{near}}/\kappa_{\text{far}}), \quad (23)$$

so that $\kappa_{\text{near}} = \kappa_{\text{far}} e^\theta$. Parameter derivatives are computed by propagating the column state and its sensitivity,

$$p' = T^\top p, \quad s' = T^\top s + T_\theta^\top p, \quad s(0) = 0, \quad (24)$$

in one augmented matrix exponential; all semigroup actions are sparse matrix-exponential actions in IEEE double precision. Time derivatives use Eq. (10); no sampled-curve maximum is used to locate the fold, which is obtained by a Powell hybrid root solve on the scaled residual pair (f_t, f_{tt}) with parameter tolerance 10^{-11} and guarded acceptance thresholds.

The numerical root is

$$t_* = 37.0749586402, \quad \theta_* = -9.6753635853, \quad (25)$$

corresponding to $\kappa_{\text{near}} = 5.755410148 \times 10^{-5}$. Residuals and margins are made dimensionless with the fold time and density: $|f_t|t_*/f = 7.1 \times 10^{-15}$, $|f_{tt}|t_*^2/f = 6.8 \times 10^{-13}$, third derivative $f_{ttt}t_*^3/f = 69.94$, and parameter transversality $f_{t\theta}t_*/f = -0.1962$. Independent continuation uses a symmetric geometric ladder of fourteen control offsets $\Delta\theta = \pm 0.001$ – 0.064 that play no role in locating the fold; the log-log fits use the six two-peak-side offsets $\Delta\theta \leq 0.032$ and give slopes 0.50095 for separation and 1.50877 for prominence, while the seven one-peak-side offsets confirm the absence of any critical pair (Fig. 2). The normal-form amplitude is tested as well: the predicted separation $2\sqrt{-2a_*/b_*} \Delta\theta^{1/2} = 5.554 \Delta\theta^{1/2}$ matches the measured separations to 0.01% at $\Delta\theta = 0.001$ and 0.7% at $\Delta\theta = 0.064$. Mass closure is 4.0×10^{-15} , and survival at $t = 5000$ is 1.3×10^{-21} .

The very small near-channel probability is reported as a limitation. The fold is derivative-certified even though the emerging feature is not a practically balanced two-peak signal. The associated synchronous discovery, Poissonized-clock comparison, and GIG approximation errors are secondary mechanism checks and are kept in the Supplemental Material [20].

B. Finite-radius two-dimensional matched-budget folds

The second family—labeled M2D-F, for matched-budget two-dimensional finite-radius—is a separate bounded realization. Two particles evolve on the obstacle-free unit square with reflecting outer boundaries. Each one-particle generator is a conservative boundary-node CTMC with central nearest-neighbor diffusion, upwind longitudinal drift, and a smooth transverse restoring drift: on the uniform grid with spacings h_x, h_y , a node jumps to its right or left neighbor at rate $D/h_x^2 + \max(\pm v_x, 0)/h_x$ and to its upper or lower neighbor at rate $D/h_y^2 + \max(\pm v_y, 0)/h_y$, with drift field $(v_x, v_y) = (v, -\gamma_\perp(y - \frac{1}{2}))$, and out-of-range jumps are omitted (reflection by omission). Reaction occurs when the particle separation is at most $a = 0.17$ and the midpoint $C_{1/2} = (X_1 + X_2)/2$ lies in one of two circular catalytic patches. The parameters are

$$(D_1, v_1) = (0.0025, 0.115), \quad (D_2, v_2) = (0.0008, 0.02),$$

with transverse confinement 1.5, starts (0, 0.5) and (0.28, 0.5), and near/far patch specifications

$$(x, r, \kappa)_{\text{near}} = (0.25, 0.18, 0.5), \\ (x, r, \kappa)_{\text{far}} = (0.72, 0.20, 15),$$

where both patch centers have $y = 0.5$. The full closed-mask convention, contact-safe initial-law construction,

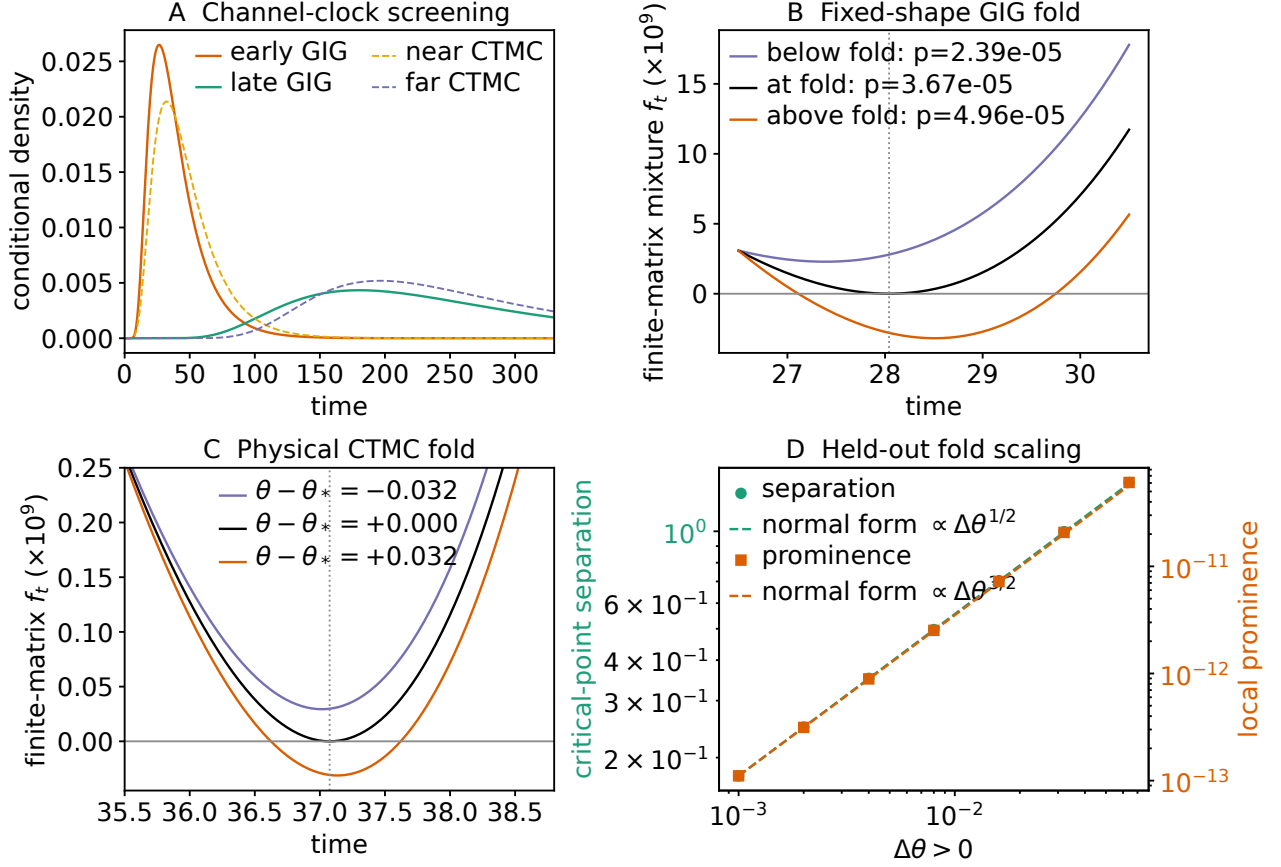


FIG. 2. Finite-CTMC fold certificate. The fold is located directly from matrix-exponential derivatives of the killed generator. Independent continuation recovers the $1/2$ separation and $3/2$ prominence exponents. The free-space GIG channel clocks shown in the figure are screening estimates, not replacements for the finite-matrix calculation; their derivation and clock-control audits are in the Supplemental Material [20]. The near-channel splitting probability at the fold is 8.95×10^{-6} , so the result establishes a mechanism rather than two comparably observable streams.

and dimensional scaling are specified in the Supplemental Material [20]. Because the tested grids have large local Péclet numbers and binary supports, they are treated as finite generators rather than a controlled SDE refinement.

On each grid, a homogeneous rate $\bar{\kappa}_h$ is recomputed so that the unweighted statewise killing sum equals that of the patterned endpoint. The reactive-tube state counts and matched rates are $(N_T, N_{\text{near}}, N_{\text{far}}, \bar{\kappa}_h) = (125, 7, 9, 1.108)$, $(349, 34, 40, 1.768)$, and $(1123, 109, 137, 1.878)$ on the 9×5 , 11×7 , and 13×9 grids, and the build aborts unless the killing sum at both path endpoints matches the budget to 2×10^{-12} . The physical path is

$$\kappa_\theta(C_{1/2}) = (1 - \theta)\bar{\kappa}_h + \theta\kappa_{\text{pat}}(C_{1/2}), \quad 0 \leq \theta \leq 1. \quad (26)$$

Thus the discrete state-sum killing budget is fixed to roundoff while both splitting weights and conditional channel shapes may vary. M2D-F is not a continuation of the endpoint family M2D-E in the Supplemental Material: transport parameters, reaction radius, starts, and per-grid budget paths differ.

Sparse generator actions give f_t, f_{tt}, f_{ttt} , and an augmented exponential gives the parameter derivatives; the fold system is solved with its analytic 2×2 Jacobian at parameter tolerance 10^{-11} , under the same normalized acceptance guards as the encounter chain. The three physical-interval folds are

$$\begin{aligned} (t_c, \theta_c)_{9 \times 5} &= (16.8093321, 0.27537300), \\ (t_c, \theta_c)_{11 \times 7} &= (18.0995323, 0.01381035), \\ (t_c, \theta_c)_{13 \times 9} &= (16.5807587, 0.25589201). \end{aligned} \quad (27)$$

The 9×5 residuals are below 3.5×10^{-17} , with $f_{ttt} = -1.34 \times 10^{-5}$ and $f_{t\theta} = 1.67 \times 10^{-4}$. The 11×7 residual pair is $(8.7 \times 10^{-19}, 3.5 \times 10^{-18})$, with $f_{ttt} = -7.81 \times 10^{-6}$ and $f_{t\theta} = 2.83 \times 10^{-4}$. The 13×9 residuals are below 2.8×10^{-17} , with $f_{ttt} = -5.96 \times 10^{-6}$ and $f_{t\theta} = 7.46 \times 10^{-5}$.

Independent continuation from eight control offsets $\mu = \theta - \theta_c \in [2 \times 10^{-4}, 0.05]$, fitted on the six offsets $\mu \leq 0.01$, gives separation/prominence exponents $(0.4989, 1.5001)$, $(0.4933, 1.4843)$, and $(0.4993, 1.5039)$. All three generators therefore satisfy the local nonde-

generate normal form. They do not form a convergent critical-control sequence: the state-count-matched θ_c values are nonmonotone and span 0.262. Replacing equal node weights in the budget by tensor-product boundary trapezoidal weights (per-axis weight $\frac{1}{2}$ on the two boundary nodes and 1 in the interior, tensorized over both particles), without changing the CTMC generators or masks, moves the folds to 0.6244, 0.2408, 0.4593, again nonmonotonically, with span 0.384. This one-factor audit shows that the finite-model fold is also sensitive to the budget measure.

Figure 3 collects the fixed-budget construction, the endpoint morphology audit, the derivative-located folds, and their independent normal-form scaling.

The two intervening grids are consistent with that caution. On 12×8 , a nondegenerate continuation root lies at $(t_c, \theta_c) = (18.26877, -0.06158)$, outside the physical path. On 10×6 , a curvature scan over $\theta \in [-0.2, 0.1]$, $t \in [0.01, 40]$ and four bounded least-squares starts (bounds $t \in [8, 25]$, $\theta \in [-0.2, 0]$) find no fold, with all starts reaching the same positive near miss $f_t = 3.8721 \times 10^{-6}$. This bounded search does not cover the interval $\theta \approx 0.25$ – 0.28 in which two of the three located folds sit; it is a bounded diagnostic, not evidence of absence. Detailed endpoint roots, budget quadrature, even-grid scans, and morphology settings are provided in the Supplemental Material [20].

V. CONTINUUM BRIDGE: THEORY, REFINEMENT, AND OFF-LATTICE VALIDATION

The finite-generator certificates of Sec. IV deliberately stopped short of any continuum claim. This section closes that gap in three steps: an exact transfer theory specifying what discrete-to-continuum fold inheritance requires; a cell-averaged, successively refined lattice family that removes the two discretization pathologies of the original grids; and a lattice-free stochastic realization of the continuum model. The combined result is sharper than an extrapolated critical value: the physical matched-budget path carries *no* interior fold in the continuum limit, and the fold responsible for the finite-grid observations sits outside the physical path, extrapolating to the admissibility edge of the redistribution family where the far-channel rate vanishes.

A. Exact transfer statements

Write $F_h(t, \theta) = f_h(t, \theta)$ for the level- h reaction density and F for a candidate continuum limit on a compact rectangle $N = [t_1, t_2] \times [\theta_1, \theta_2]$ with $t_1 > 0$.

a. Jet convergence from sectorial resolvent convergence. Let $A_h(\theta) = L_h - \text{diag } k_{h,\theta}$ be the killed generators, with $f_h(t, \theta) = \langle k_{h,\theta}, e^{A_h(\theta)t} p_h \rangle$, and let $A(\theta)$ be the continuum killed operator. Assume: (H1) the

family $\{A_h(\theta)\}$ is uniformly sectorial—there are $M \geq 1$, $\omega_0 \in \mathbb{R}$, $\delta \in (0, \pi/2)$, independent of h and θ , with $\|(z - A_h(\theta))^{-1}\| \leq M/|z - \omega_0|$ for $|\arg(z - \omega_0)| < \frac{\pi}{2} + \delta$; (H1') $A(\theta)$ is sectorial on the same sector with the analogous bound, uniformly in θ ; the continuum family is affine along the same path [Eq. (26)], $A(\theta) = A(0) - \theta(k_{\text{pat}} - \bar{k})$ acting by multiplication, with θ -differentiable resolvent; the pairing $\langle k_\theta, \cdot \rangle$ with p is well defined uniformly in θ ; and $f(t, \theta) = \langle k_\theta, e^{A(\theta)t} p \rangle$ admits the analytic-semigroup representation below, so that $\partial_\theta \partial_t^j f$ has the same two-term Dunford form as its discrete counterpart. Fix $\delta' \in (0, \delta)$, $\rho > 0$, and the *unbounded* sectorial contour $\Gamma = \{\omega_0 + re^{\pm i(\pi/2 + \delta')} : r \geq \rho\} \cup \{\omega_0 + \rho e^{i\varphi} : |\varphi| \leq \frac{\pi}{2} + \delta'\}$, oriented from the lower ray to the upper ray (counterclockwise about the spectrum). For $t \geq t_1 > 0$ the Dunford representation

$$\partial_t^j f_h(t, \theta) = \frac{1}{2\pi i} \int_\Gamma z^j e^{zt} \langle k_{h,\theta}, (z - A_h(\theta))^{-1} p_h \rangle dz, \quad j \leq 3,$$

holds exactly for every h (and, by (H1'), for the continuum pair). Define the scalar contour defects

$$\eta_h = \sup_\theta \sup_{z \in \Gamma} |\langle k_{h,\theta}, (z - A_h(\theta))^{-1} p_h \rangle - \langle k_\theta, (z - A(\theta))^{-1} p \rangle|,$$

and, using $\partial_\theta(z - A_h)^{-1} = -(z - A_h)^{-1} \text{diag}(k_{h,\text{pat}} - \bar{k}_h)(z - A_h)^{-1}$ (the path is affine, $A_h(\theta) = A_h(0) - \theta \text{diag}(k_{h,\text{pat}} - \bar{k}_h)$), the mixed defect η'_h as the same supremum for the two-term derivative form $\langle k_{h,\text{pat}} - \bar{k}_h, (z - A_h)^{-1} p_h \rangle - \langle k_{h,\theta}, (z - A_h)^{-1} \text{diag}(k_{h,\text{pat}} - \bar{k}_h)(z - A_h)^{-1} p_h \rangle$ against its continuum counterpart. Then, for $j \leq 3$, $l \in \{0, 1\}$, $j + l \leq 3$,

$$\sup_N |\partial_\theta^l \partial_t^j (f_h - f)| \leq C_\Gamma^{(j)}(t_1, t_2) (\eta_h + l \eta'_h), \quad C_\Gamma^{(j)} = \frac{1}{2\pi} \int_\Gamma \max(1$$

finite because on the rays $\text{Re } z = \omega_0 - r \sin \delta'$, so the integrand decays like $r^j e^{-t_1 r \sin \delta'}$; the constant grows like $t_1^{-(j+1)}$ as $t_1 \downarrow 0$, which is why the strictly positive window is load-bearing. Hypotheses (H1), (H1') and the decay of η_h, η'_h are *not* proved here for the upwind or exponentially fitted schemes; they are the precise analytic content that the observed ladder rates instantiate numerically.

b. Quantitative fold transfer. Let F, F_h have continuous $\partial_t^j F$ ($j \leq 3$) and $\partial_\theta \partial_t^j F$ ($j \leq 2$) on N . Assume F has an interior fold at $q_* = (t_*, \theta_*)$: $F_t(q_*) = F_{tt}(q_*) = 0$, with margins $|F_{ttt}| \geq \mu_3 > 0$ and $|F_{t\theta}| \geq \mu_\theta > 0$ on N . Let Λ bound the sup norms $\|F_{t\theta}\|_N, \|F_{ttt}\|_N, \|F_{tt\theta}\|_N$, and let ω (part of the data) be a common nondecreasing modulus of continuity for these three functions on N ; write $\hat{\omega}(r) = \max(\omega(r), \sqrt{2}\Lambda r)$ for the augmented modulus that also controls the F_{tt} entry. Define

$$\varepsilon_h = \max_{j=1,2} \|\partial_t^j (F_h - F)\|_N, \quad \varepsilon'_h = \max \left(\|\partial_t^3 (F_h - F)\|_N, \max_{j \in \{1,2\}} \|\partial_\theta \partial_t^j (F_h - F)\|_N \right)$$

This is the C^3 -versus- C^2 gap: two time derivatives control the location of the critical pair, and the third controls its nondegeneracy. Let $H = (F_t, F_{tt})$ with Jacobian

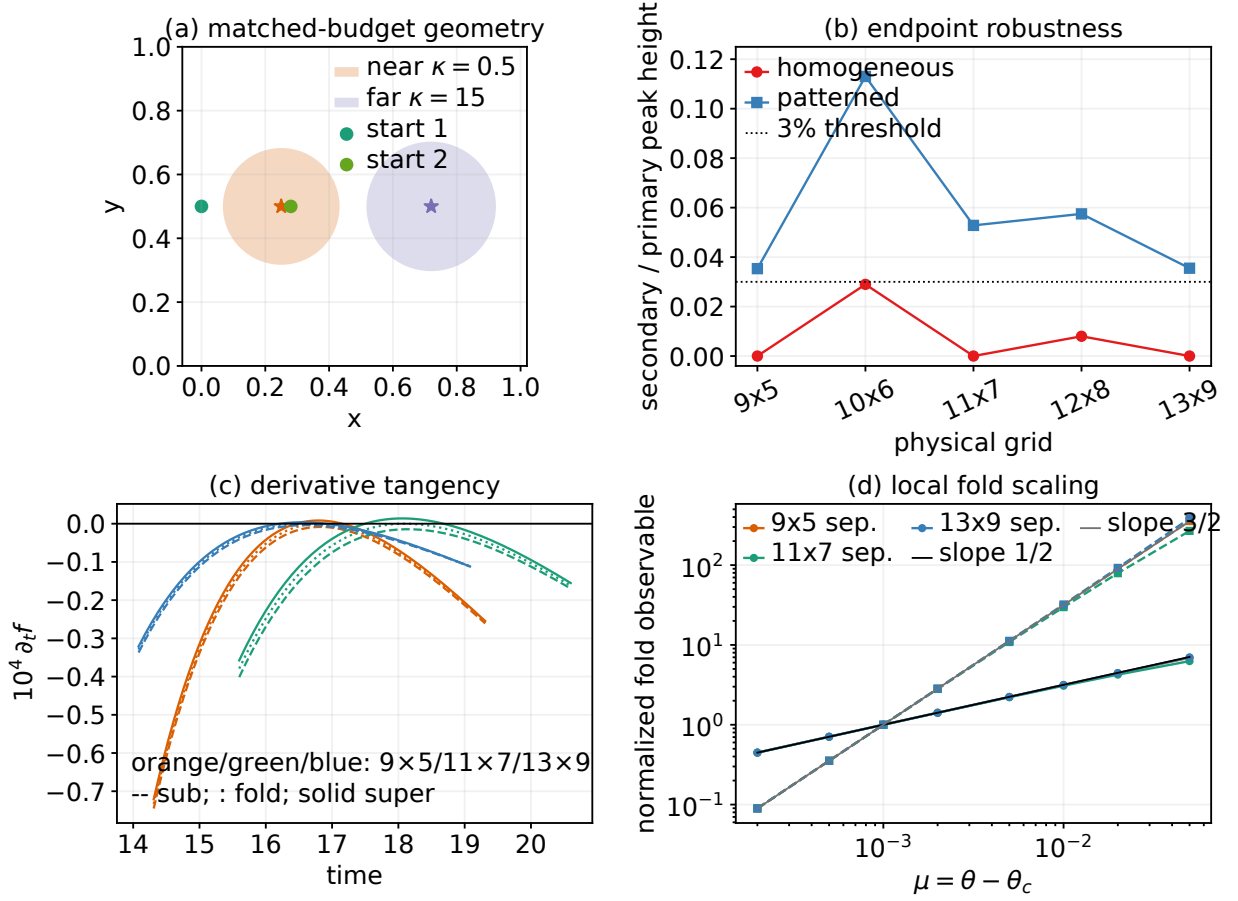


FIG. 3. Finite-radius two-dimensional matched-budget (M2D-F) fold certificate. (a) Spatial redistribution keeps the statewise killing sum fixed on each grid. (b) The endpoint audit distinguishes resolved secondary peaks from reported subthreshold ripples. (c) Analytic derivatives locate a tangency and its two-sided continuations on three finite grids. (d) Each grid recovers square-root separation and 3/2 prominence. The critical controls 0.275, 0.014, 0.256 are explicitly nonmonotone and move under a second budget measure; they are not estimates of a continuum fold.

$$J = \begin{pmatrix} F_{tt} & F_{t\theta} \\ F_{ttt} & F_{tt\theta} \end{pmatrix}, \text{ so } \det J(q_*) = -F_{t\theta}(q_*)F_{ttt}(q_*) \text{ and}$$

$$\sigma := 1/\|J(q_*)^{-1}\| \geq \frac{\mu_3\mu_\theta}{\sqrt{\mu_\theta^2 + 2\Lambda^2}} > 0.$$

Let $r_0 = \min(r_\omega, \text{dist}(q_*, \partial N))$, where r_ω is any radius with $\hat{\omega}(r_\omega) \leq \sigma/16$ (in particular $r_\omega \leq \sigma/(16\sqrt{2}\Lambda)$); so r_0 is determined by $(\mu_3, \mu_\theta, \Lambda, \omega)$ and the position of q_* in N), and set

$$\delta_0 = \min\left(\frac{\sigma}{32}, \frac{\mu_3}{2}, \frac{\mu_\theta}{2}, \frac{5\sigma r_0}{8\sqrt{2}}\right).$$

If $\bar{\varepsilon}_h < \delta_0$, then F_h has exactly one solution q_h of $F_{h,t} = F_{h,tt} = 0$ in $B(q_*, r_0)$; it is a nondegenerate fold of F_h , with margins $|F_{h,ttt}| \geq \mu_3 - \bar{\varepsilon}_h > 0$ and $|F_{h,t\theta}| \geq \mu_\theta - \bar{\varepsilon}_h > 0$ on all of N ; and

$$\|q_h - q_*\| \leq 2\sqrt{2}\|J(q_*)^{-1}\|\varepsilon_h.$$

Converse (ladder version). Suppose each F_h has a fold $q_h \in N$ with $\text{dist}(q_h, \partial N) \geq d > 0$ uniformly, the mar-

gins $|F_{h,ttt}| \geq \mu_3$, $|F_{h,t\theta}| \geq \mu_\theta$ hold on N , the functions $F_{h,t\theta}, F_{h,ttt}, F_{h,tt\theta}$ are bounded by Λ with a common non-decreasing modulus ω uniformly in h , and the jets $\partial_t^j F_h$ ($j \leq 3$), $\partial_\theta \partial_t^j F_h$ ($j \in \{1, 2\}$) converge uniformly on N to those of F . Then for all sufficiently fine h the limit F has a nondegenerate fold $\bar{q}$ with

$$\|\bar{q} - q_h\| \leq 2\sqrt{2} \frac{\sqrt{\mu_\theta^2 + 2\Lambda^2}}{\mu_3\mu_\theta} \varepsilon_h,$$

and $\bar{q}$ is the *only* fold of F in N : the N -wide margins force F_{ttt} and $F_{t\theta}$ to keep constant signs on the connected rectangle, so $F_t(\cdot, \theta)$ has a one-signed extremum with value zero at any fold and strict monotonicity of F_t in θ excludes a second fold at any other θ . The proof [20] is a simplified-Newton contraction seeded at the fold; no Lipschitz control of the defect $F_h - F$ is required.

c. Remark (a posteriori interpretation). Three statements of decreasing strength organize how the ladder is read. (R1) *Rigorous-conditional*: if the jets of F_h converge uniformly on N to some F and the

discrete folds, margins, and interior distances satisfy the converse hypotheses uniformly along the ladder, then F has a nondegenerate fold and every located discrete fold lies within $2\sqrt{2}\sqrt{\mu_\theta^2 + 2\Lambda^2}(\mu_3\mu_\theta)^{-1}\varepsilon_h$ of it. (R2) *Falsification bound*: if two discretizations both satisfy the transfer hypotheses toward the same limit and each located fold is the unique one of the theorem’s ball (both schemes track the same fold branch), their located critical controls obey $|\theta_c^{(1)}(h) - \theta_c^{(2)}(h)| \leq 2\sqrt{2}\|J(q_*)^{-1}\|(\varepsilon_h^{(1)} + \varepsilon_h^{(2)})$; a large cross-scheme gap therefore falsifies joint convergence at the claimed rates, while a small gap is necessary but not sufficient—error components shared by both schemes (the cell-averaged masks and the budget rule are common to both) cancel to leading order in the gap and are not tested by it. (R3) *Error estimate under an asymptotic-regime assumption*: if the level-to-level jet increments $d_k = \max_{j \leq 2} \|\partial_t^j(F_{h_{k+1}} - F_{h_k}^{\text{lad}})\|_N$ obey the geometric domination $\bar{d}_{m+1} \leq \varrho \bar{d}_m$ for all $m \geq k$ with a fixed $\varrho < 1$ —the observed level ratios certify this only on the computed levels, so ϱ should dominate, not average, them, and the extension to uncomputed levels is an assumption—then the telescoping bound $\tilde{\varepsilon}_{h_k} := \max_{j \leq 2} \|\partial_t^j(F_{h_k} - F_\infty^{\text{lad}})\|_N \leq d_k/(1 - \varrho)$ controls the distance to the ladder’s own jet limit F_∞^{lad} ; its identification with the continuum F is exactly the (H1)/(H1’) content. Richardson extrapolants of $\theta_c(h)$ are reported as estimates only: no convergence rate is claimed for them unless the fitted leading coefficient is nonzero, the observed order is stable across consecutive level triples, and a strengthened remainder $O(h^{p+s})$, $s > 0$, is assumed. The cross-scheme gap of the two extrapolants is reported alongside them as a conservative uncertainty indicator.

B. Cell-averaged refined family

The refined family keeps every physical parameter of Sec. IV B (transport, starts, contact radius, patch geometry and rates, matched budget) and changes only the discretization, removing its two known pathologies. First, the binary masks are replaced by cell averages: for every product state the fraction of its four-dimensional product cell inside the killing region $\{|x_1 - x_2| < a\} \cap \{C_{1/2} \in \text{patch}\}$ is computed by tensorized midpoint subsampling with four points per axis per particle cell ($4^4 = 256$ subsamples per product cell), so the killing field varies smoothly as the grid moves across patch boundaries instead of aliasing. Second, the grids are square-celled, $n \times n$ nodes with $h = 1/(n - 1)$ per axis, so no aspect anisotropy enters the ladder. The budget is matched by tensor-product boundary control-volume (trapezoidal) weights, the per-grid homogeneous rate is recomputed so that the weighted killing sum is θ -invariant to 2×10^{-12} relative (verified at both endpoints), and the per-grid budget converges to the continuum tube integral (absolute gap 7.3×10^{-3} at $n = 9$, 3.6×10^{-3} at $n = 13$, $5.8 \times$

TABLE II. Cell-averaged square-cell refinement ladder: located fold coordinates for the two transport discretizations (upwind, UW; Scharfetter–Gummel, SG). Every accepted root has normalized residuals at most 2.4×10^{-12} and nondegenerate normalized margins [20]. All folds from $n = 17$ on lie outside the physical path $\theta \in [0, 1]$; the pre-asymptotic levels $n \leq 13$ are shown for completeness (the UW solver locks onto a different root branch with $t_c \approx 10.5$, and the $n = 9$ SG root at $\theta_c = 0.215$ is a residual coarse-cell root that does not survive refinement).

n	θ_c^{UW}	θ_c^{SG}	t_c^{UW}	t_c^{SG}
9	−0.1837	0.2148	10.387	18.607
13	−0.1758	−0.1178	10.578	19.327
17	−0.0431	−0.1386	17.952	18.293
21	−0.1000	−0.1458	18.273	17.796
25	−0.1181	−0.1507	18.301	17.442
29	−0.1297	−0.1553	18.228	16.977
33	−0.1370	−0.1578	18.174	16.811

10^{-4} at $n = 17$, and bounded by 2.0×10^{-3} with alternating sign at every finer level). Two transport discretizations are run over the same ladder: the upwind rates of Sec. IV B and Scharfetter–Gummel exponential fitting [23], with per-edge Péclet number $\text{Pe} = v_{\text{edge}}h/D$ and rates $(D/h^2)\text{Pe}/(1 - e^{-\text{Pe}})$ forward, $(D/h^2)\text{Pe}/(e^{\text{Pe}} - 1)$ backward, which are positive for all Pe and reduce to central differencing as $\text{Pe} \rightarrow 0$. Folds are located exactly as before (augmented sensitivities, analytic Jacobian, tolerance 10^{-11} ; first-derivative residuals below 3.5×10^{-13} and second-derivative residuals below 2.5×10^{-12} at every accepted root).

Table II and Fig. 4(a) report the ladder. Three features are decisive. First, from $n = 17$ on, *both* discretizations place the fold at negative control, outside the physical path $\theta \in [0, 1]$, with nondegenerate normalized margins throughout. Second, the two sequences approach each other as the grid refines: the cross-discretization gap in θ_c falls from 0.095 at $n = 17$ to 0.046 at $n = 21$, 0.033 at $n = 25$, 0.026 at $n = 29$, and 0.021 at $n = 33$. Third, the coarse levels $n \leq 13$ are pre-asymptotic: the upwind solver locks onto a different root branch (fold times near $t \approx 10.5$ instead of $t \approx 17$ –18), and the $n = 9$ Scharfetter–Gummel fold is the single physical-interval exception ($\theta_c = 0.215$) that disappears under its own refinement; at $n = 9$ a further 7.1% of the initial mass lies inside the contact tube (exactly zero at every finer level), compounding the coarse-cell error at that single level. Restricting to the same-branch levels $n \geq 17$, first-order Richardson extrapolation on the finest level pair gives $\theta_* \approx -0.19$ (upwind) and $\theta_* \approx -0.18$ (Scharfetter–Gummel); the observed order is not yet stable across level triples, so, per the a posteriori remark of Sec. V A, these extrapolants are reported as estimates with the cross-scheme gap as a conservative uncertainty indicator, and no convergence rate is claimed for them. They are to

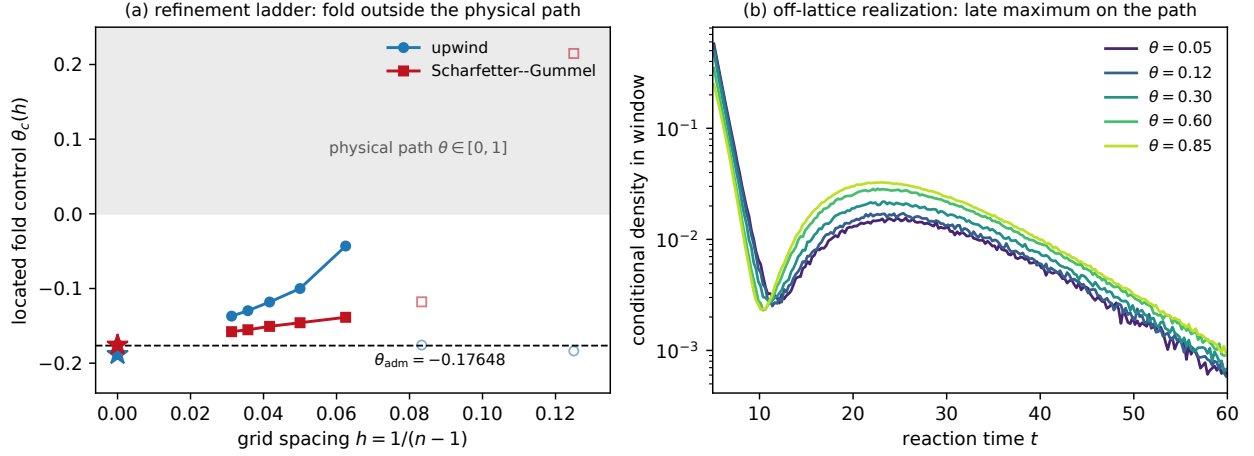


FIG. 4. Continuum bridge. (a) Located fold control $\theta_c(h)$ for the cell-averaged square-cell ladder under both transport discretizations (filled markers: same-branch levels $n \geq 17$; open markers: the pre-asymptotic levels $n \leq 13$ discussed in the text). Every same-branch root lies below the physical path (shaded); the stars at $h = 0$ are the first-order Richardson estimates, which sit at the admissibility edge θ_{adm} [Eq. (28)] (dashed) within their uncertainty. (b) Off-lattice reaction-time densities (2×10^7 walkers per control, conditioned on the window $t \in [5, 60]$): the valley-late-maximum structure is present at every sampled control and strengthens monotonically with θ .

be compared with the admissibility edge of the matched-budget family: the path rate [Eq. (26)] on the far patch, $(1 - \theta)\bar{\kappa} + 15\theta$, first becomes negative at

$$\theta_{\text{adm}} = -\frac{\bar{\kappa}}{15 - \bar{\kappa}} = -0.17648 \dots, \quad \bar{\kappa} = 2.2501572 \dots, \quad (28)$$

so the extrapolated fold sits at (or within the extrapolation uncertainty of) the boundary at which the far channel shuts off entirely. What the ladder does establish, at every level $n \geq 17$ and for both discretizations, is that the located fold lies outside the physical path: the three binary-mask folds of Sec. IV B do not persist under refinement, and their nonmonotone, budget-measure-sensitive critical controls are explained as artifacts of the coarse binary-mask, equal-weight discretization they were computed with.

C. Off-lattice stochastic validation

A lattice-free realization removes the remaining dependence on any grid. Two particles obey the continuum model directly (Euler-Maruyama, $dX_i = v_i dt + \sqrt{2D_i} dW$, transverse Ornstein-Uhlenbeck restoring drift, reflecting walls, $dt = 2 \times 10^{-4}$), and while $|X_1 - X_2| < a$ the pair is killed at rate $\kappa_\theta(C_{1/2})$ with per-step probability $1 - e^{-\kappa dt}$. The continuum homogeneous matched rate $\bar{\kappa} = 2.2501572$ reproduces the exact tube-integral reference to all printed digits and is cross-checked by a 10^8 -sample Monte-Carlo volume estimate to 0.12%. Consistency checks: zero kills at $\kappa = 0$ over 2.5×10^7 contact steps; per-step kill probabilities within $[0, 1]$; halving dt at 2×10^5 walkers ($\theta = 0.12$ and 0.60)

changes the windowed histograms by amounts consistent with pure counting noise (per-bin $|z|$ averaging 0.72–0.75, $\max |z| \leq 3.1$ over 50–55 bins).

With 2×10^7 walkers per control value, the reaction-time density shows a significant late second maximum throughout the sampled physical path: at $\theta = 0.05, 0.12, 0.30$ the smoothed histograms give second maxima at $t = 24.6, 23.4, 22.9$ with Poisson significances $12.6\sigma, 14.0\sigma, 18.7\sigma$ and $0.8\text{--}1.8 \times 10^5$ events in the diagnostic window $t \in [10, 30]$; at $\theta = 0.60$ the second maximum reaches 31.8σ with 5.1×10^5 window events, and at $\theta = 0.85$ it reaches 52.3σ at $t = 22.6$ with 1.4×10^6 window events. The late-peak prominence *increases* monotonically with θ over the sampled range: patterned redistribution strengthens the far-channel peak, and nowhere on the sampled physical path does the maximum-minimum pair annihilate. Figure 4(b) shows the windowed densities. The off-lattice model therefore agrees with the refined ladder on both counts: the physical family is uniformly two-peaked, and the fold that the finite-grid family kept locating inside $[0, 1]$ is not a continuum feature of the physical path.

D. Interpretation

Three independent lines—the transfer theory, the two-discretization refinement ladder, and the lattice-free realization—assemble into one statement: for this geometry the matched-budget redistribution path possesses no interior continuum fold; the organizing fold of the finite-window morphology lies just outside the path, at negative control near θ_{adm} [Eq. (28)], where the admissible family terminates because the far-patch rate reaches zero.

The finite-generator certificates of Sec. IV remain exactly what they were declared to be: correct statements about their own generators, whose critical controls are now shown—rather than merely suspected—not to survive refinement. The practical lesson for reaction-field design is correspondingly sharpened: on this path, spatial redistribution at fixed budget *tunes* the relative weight of the two arrival-time scales but cannot merge them; the merger happens only at the edge of admissibility, in the limit of complete far-channel shutdown.

VI. DISCUSSION AND SCOPE

The finite-model calculation separates three questions. Transport determines which temporal scales are available. The initial and reaction observables determine the signs and amplitudes of their spectral residues. Finally, a declared feasible reaction-field perturbation determines whether the critical-point equation can be crossed. Patch count alone addresses none of these questions: a support-rank-one counterexample can already be bimodal, whereas a residue sequence with three sign changes need not have three positive derivative roots.

The Green reduction and sensitivity calculus play complementary roles. $\mathcal{G}(s)$ reduces transport to the reaction support, while Eqs. (16)–(19) operate directly on the time-domain fold equation. A killed pole can be dark or cancel from a selected observable, and a density fold is not a spectral exceptional point. The spectral sign count is therefore used only to falsify an impossible finite reversible mode count, never to replace direct time-derivative root calculations.

The finite encounter CTMC confirms the derivative chain but has a very small minority channel. The M2D-F result more directly tests spatial redistribution at a fixed per-grid budget, yet its grid and budget-measure sensitivity prevents a continuum interpretation. The supporting Supplemental Material shows additionally that spatial patterning can promote an existing subthreshold maximum across an operational resolution threshold in a separate endpoint family, but is not necessary for bimodality; a uniformly reactive finite model provides a counterexample [20]. It also records catalytic-coordinate sensitivity and underresolved three-patch finite-grid examples. Those controls restrict causal language rather than broaden the principal claim.

The continuum bridge is settled in Sec. V. The C^3 -versus- C^2 distinction that governs fold transfer is made exact by the transfer theorem there, and the cell-averaged, two-discretization refinement it prescribes returns a negative verdict for this geometry: no interior fold survives on the physical path, and the finite-grid critical controls are artifacts of the coarse binary-mask discretization (Secs. VB and VC).

Other limits remain explicit. The projected optimum is local and ignores finite-amplitude nonnegativity, box constraints, binary supports, and nonsmooth patch motion.

The Green identity is a right-half-plane volume-reaction statement under its operator hypotheses; an unqualified Robin trace analogue and continuum meromorphic continuation are not supplied. The original 2D grids use upwind transport and binary masks; the cell-averaged ladder and Brownian realization of Sec. V quantify exactly what those choices affected. The sectorial and jet hypotheses of Sec. VA are instantiated numerically rather than proved for these schemes, the ladder extrapolants carry no claimed convergence rate, and the off-lattice family samples five control values. The chosen discrete state-sum budget does not equalize pathwise hazards, splitting probabilities, or mean lifetimes. Capacity and GIG calculations in the Supplemental Material are calibration and screening layers, not centre-patterned modality theorems.

VII. CONCLUSIONS

For a declared finite encounter generator, the observable spectral residues give a necessary sign-variation gate, and the Fréchet–Duhamel derivative gives the exact local response of the critical-point equation to the killing field. Projection onto a budget tangent space then identifies the steepest unit feasible redistribution in a chosen metric. These statements link transport, reaction observability, and admissible control without equating the number of catalytic patches with the number of modes.

Two numerical certificates exercise the calculus. A finite two-reactant CTMC has a nondegenerate density fold with direct residual and transversality checks and the expected continuation exponents. A separate finite-radius M2D-F path has analogous folds on three finite grids under an exactly fixed per-grid killing sum. Their critical controls are nonmonotone and budget-measure sensitive, and the continuum study of Sec. V explains why: under cell-averaged refinement with two independent transport discretizations the located fold leaves the physical path, and the lattice-free realization confirms a two-peaked density at every sampled control. The finite-generator certificates therefore stand as declared, and the continuum statement this article adds is a negative one with a sharp boundary mechanism: at fixed budget, redistribution on this path tunes but cannot merge the two arrival-time scales; merger occurs only at the admissibility edge of complete far-channel shutdown.

DATA AVAILABILITY

The source code, numerical tables, vector figures, and machine-readable provenance manifests that reproduce every reported number are available from the author upon reasonable request, and will accompany the published article as a public, versioned archive.

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